Introduction to Theoretical Computer Science

CS2490 - Session 21: Grammars and regular languages

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Today, we see a 4th way: Regular Grammars.

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Using these rules, we can generate sentences like “the cat sits on the mat” or “my dog barks loudly”.

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We write $G = (\Sigma, V, S, P)$.

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3. The grammar $G = (\{a, b, c\}, \{S, B, C\}, S, P)$ where P contains:

$$S \to aSBC \mid aBC, \quad CB \to BC, \quad aB \to ab, \quad bB \to bb, \quad bC \to bc, \quad cC \to cc$$

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- We repeat this until we get a word containing only terminal symbols (i.e., symbols from Σ).
- The language $\mathrm{L}(G)$ is the set of all words we can generate this way.

Grammars generate languages (cont'd)

Formally, write $u \Rightarrow v$ if $v \in (V \cup \Sigma)^*$ can be obtained from $u \in (V \cup \Sigma)^*$ by applying a production rule $x \to y$ in P

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Then $\mathrm{L}(G)$ is the language of all words (without variables) derivable from S:

$$\mathrm{L}(G) = \{w \in \Sigma^* \mid S \Rightarrow^* w\}$$

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Generally, a derivation is of the form: $S \overset{P_1}{\Rightarrow} aSb \overset{P_1}{\Rightarrow} \ldots \overset{P_2}{\Rightarrow} a^n b^n$ for some $n \geq 0$.
So $\mathrm{L}(G) = \{a^n b^n \mid n \geq 0\}$.

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Consider the grammar $G = (\{0,1\}, \{S, A, B\}, S, P)$ where P contains the rules:

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- $S \Rightarrow 0A \Rightarrow 00AA \Rightarrow 001SA \Rightarrow 0010AA \Rightarrow 0010A1S \Rightarrow 0010A1 \Rightarrow 00101S1 \Rightarrow 001011$

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Here, $\mathrm{L}(G)$ is a bit more complicated to determine. Take a guess!

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In each rule: the left-hand side is a single variable; and the right-hand side is a terminal word possibly followed by a single variable (not necessarily the same as the left-hand side).

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A derivation of a word from S is necessarily of the form:

$$S \Rightarrow 0S \Rightarrow 00S \Rightarrow \cdots \Rightarrow 0^n S \Rightarrow 0^n 1A \Rightarrow 0^n 11A \Rightarrow \cdots \Rightarrow 0^n 11^m A \Rightarrow 0^n 11^m$$

for some $n, m \geq 0$.

Example

Consider the grammar $G = (\{0,1\}, \{S, A\}, S, P)$ where P contains the rules:

$$S \to 0S, \quad S \to 1A, \quad A \to 1A, \quad A \to \varepsilon$$

This is a regular grammar: all rules are of the form Variable $\to$ word Variable or Variable $\to$ word.

Question: What is $\mathrm{L}(G)$?

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for some $n, m \geq 0$. That is, $\mathrm{L}(G) = \mathrm{L}(0^*11^*)$.

Example

Consider the grammar $G = (\{a, b\}, \{Ee, Eo, Oe, Oo\}, Ee, P)$ where P contains the rules:

$$Ee \to \varepsilon \mid aOe \mid bEo, \quad Eo \to aOo \mid bEe, \quad Oe \to aEe \mid bOo, \quad Oo \to aEo \mid bOe.$$

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